

Periodized fascia training for speed, power, and injury resilience

Bill Parisi and Johnathon Allen

In the 1990s, the common assumption was that speed was not something you could dramatically improve through training. It was a genetic gift. You were either born to be fast or you weren't. It was believed that if you had more fast-twitch type II muscle fibers and a leaner frame, you were inevitably going to be faster. While there's obviously some inherent truth to that, it turns out that this is not the end of the story. Our understanding of speed has evolved significantly over the past few decades to give us new insights. Not only have we proven that human speed can be significantly enhanced through targeted training, we have a much better understanding of how and why (Clark and Weyand, 2014). In addition to demonstrating that the fascial system is both adaptable and trainable, modern research and imaging technology suggest it may play a significant role in the body's ability to generate speed in all of its many forms (Schleip and Müller, 2012) – from running a sub-10-second 100-meter dash, to throwing a 100-mph fast ball, and delivering a lightning-fast knockout punch. In all these examples, speed comes from multiple anatomical systems working together in a highly coordinated unison, enabled by the body's integrated myofascial web. When I was in high school, I went to Finland to train with the top javelin throwers and coaches in the world. While I was there, I was exposed to advanced functional training and medicine ball techniques not previously seen in the US. My belief is that those techniques optimize the adaptive remodeling and elastic energy-storage properties of the body's tendons

and fascial system. This functional approach to three-dimensional, fascia-focused training allowed me to learn that throwing things is one of the most complete forms of athleticism. To be a great thrower, you need powerful legs and a stiff, super-strong core. Tremendous amounts of force are channeled from the lower body through your trunk, torso, shoulder, arm, elbow, wrist, and fingers. It's a whole-body experience (Naito et al, 2011) (see Chapter 33).

Fascia loves variability – variability in load, variability in movement, and variability of vectors (Zügel et al, 2018). And the sport of basketball is a three-dimensional game with constantly changing variables happening on multiple planes with submaximal loads and explosive movements that engage the entire body from foot to finger in a bullet-fast chain reaction of proprioception and elasticity. Spend three-plus hours a day playing basketball from a very young age (and avoid injury) and you will likely develop thicker Achilles' tendons with powerful dynamic recoil properties that give you kangaroo-like abilities for jumping, accelerating, and rapidly changing direction (Wiesinger, 2017). So, with this in mind, the question I ask is: How do we create a fascia-aware curriculum backed by research that any athlete or coach can use to improve athletic performance and injury resilience?

Fascia training 101

According to Davis's Law, fascial tissue continuously remodels itself in triple-helix strands of

Chapter 48

collagen produced by thousands of tiny fibroblast cells that organize along the lines of load and stress in the body's extra-cellular matrix using a biological process called mechanotransduction (Ingber, 1998). Essentially, the net of your body-wide fascia system is constantly remodeling itself based on movements, loading patterns, diet, and lifestyle. If you sit hunched over at a desk all day, you will probably develop densely matted concentrations of collagen in the myofascial layers of your shoulders and your upper and lower back where the stress and load are concentrated. Although this process has not yet been proven to occur, animal studies show substantial disorganization of the muscular connective tissue (perimysium, epimysium) after sustained immobilization (Järvinen and Lehto, 1993). Therefore, it may be argued that without a regular program of variable movements moving those fibers and pumping viscous fluid into those matted tissues, they will eventually harden into dense adhesions that limit your mobility. On the other hand, if you spend your days working on a farm or an Alaskan tuna boat struggling against gravity under constantly variable loads happening on multiple planes of motion that challenge your body's kinetic chain in frequently awkward positions (and avoid injury), it seems probable that you will develop a resilient three-dimensional fascia system that in conjunction with thick, powerful tendons – can give you the ability to generate tremendous amounts of force despite having a leaner/smaller frame than someone who bulks up at the gym. Science is showing us that this enhanced capability for power generation comes from the elastic amplification properties of tendons and fascia tissue and a pulsing neurological chain reaction of co-contractions happening across multiple myofascial structures that combine to create a finely-tuned balance of stability and mobility (Maas and Sandercock,

2010). If you doubt this assertion, challenge a farm kid or deep-sea fisherman to a wrestling match and see how it goes. But this takes us back to the million-dollar question: How do we harness this biological dynamic to create functional athletic training programs that improve speed, power, and injury resilience?

Fascial tissue and its behavior are based on its biological structure. If we understand the basic structure of fascia and how it behaves, we will understand how to train it. Observed simply, fascia is made up of cells, fibers, the extra cellular matrix, and water (see Fig.1.3). When it comes to the cells within fascia, we've got fibroblasts, mast cells, adipose cells, and macrophages. And all of them do very specific things. During the tissue remodeling phase of healing, macrophage activity and collagenase enzymes clean up the old tissue so we can lay down the new. Fibroblasts are the cells that lay down the new. Fibroblasts build fascia just like osteoblasts build bone. They line up along the lines of stress – or 'load paths' – and produce the protein fibers of collagen, elastin, and reticulin. Since farm kids and deep sea fishermen stress their bodies in all sorts of different lengths, angles, and positions, they end up laying down an omnidirectional latticework of protein fibers throughout their bodies that is driven by the wide variety of loading patterns happening along the many different lines of stress they are subjected to.

Importantly, though, fascia tissue remodeling is rate/load specific (Bohm et al., 2014). Elastic energy is stored and released by tendons and fascia tissue very quickly – in less than 1.2 seconds (Kawakami et al., 2002). If a force lasts longer than that, the plasticity properties of fascia will adjust to accommodate to the load. This means that training tendons and fascia tissue for more elastic explosiveness requires short, cyclic, quickly

Periodized fascia training for speed, power, and injury resilience

repeated motions, like bouncing, jumping rope, or running on the balls of your feet (as opposed to slower-contraction cycles, like bicycling or rowing). Another key aspect in this dynamic is that the Type I collagen fibers that store and release elastic energy need to be connected and glued down in the extracellular matrix. They're glued down by a fluid called ground substance, which contains soluble carbohydrate polymers. These polymers take these protein fibers and glue them together omnidirectionally. The other key element in fascia tissue is water. We have either bound water or unbound water in our tissues. Unbound water consists of free-floating H_2O molecules that are not bonded with each other. Bound water is where H_2O molecules are bonded with each other and a hydrophilic surface in their vicinity. When that happens there is strong evidence to suggest that fascia becomes stiffer and more dynamic because it is more resistant to compression (Pollack, 2001). If you're an athlete, and you've got stiffer, more dynamic fascia tissue, that means you're wearing a kind of internal weightlifting compression suit that is tighter and stretchier. This means we want the water in fascia to be bound. This can be achieved through specific movement techniques, stretching, foam rolling, mechanical manipulation, eating fresh food, drinking plenty of water, alongside other strategies.

Vector variability

One of the first things to explore when designing a periodized training program is anatomical adaptation and the importance of vector variability in preparing the body for increased levels of exertion. The basic periodization premise is that the preparation phase of anatomical adaptation leads to strength, which leads to power (i.e. the expression of strength), which leads to speed, agility, and quickness. Traditionally,

weight, rep schemes, tempos, and rest periods were the only variables in the anatomical adaptation phase of a periodized training program. Typically, weights would be used to do mostly linear, sagittal-plane movements in the gym with higher reps for anatomical adaptation, lower reps with heavier weights for strength, and even lower reps using lighter weights and faster bar movements for power. However, in a fascia-aware approach to periodization, the process of anatomical adaptation is not just a matter of repeating the same movement patterns over and over with the same tempos along the same lines of stress. If it's true that fascia likes variability, then a fascia-aware approach to anatomical adaptation should include a variety of loads, vectors, and angles that stimulate omnidirectional tissue remodeling from the outset. This is because fibroblasts will lay down collagen in a more balanced latticework of fibers that provides increased shape stability, greater elastic recoil capability, and resistance to deformation which helps improve injury resilience. If tissues in the ankles, knees, hip, trunk, arms, and shoulders are exposed to different loads at different angles during the anatomical adaptation phase, the power of mechanotransduction for omnidirectional tissue remodeling across the entire body can be harnessed. This makes vector variability a key ingredient in anatomical adaptation. It is important to understand that both linear track athletes and multidirectional field and court sport athletes will benefit from a more dynamic multidirectional stretch/recoil capacity of the fascia system. An important part of linear sprinting is the ability to maintain joint and core stiffness in the transverse plane. And it is likely that inexperienced sprinters often over-rotate or leak energy from their core and joint systems because they do not have a well-balanced fascia system and strong core.

Chapter 48



Figure 48.1 A-D Box Pattern

The box pattern drill provides vector variability. It activates the core and fascia connections in a whole-body chain reaction along the frontal plane. Work with any loaded movement tool such as a ViPR PRO (pictured), medicine ball, or kettlebell to achieve vector variability using the external load and gravity as drivers. 1. Start by holding the load in an athletic stance. 2. Raise the load overhead. 3. Rotate the load vertically while lunging to the right. 4. Return to the overhead position. 5. Rotate the load vertically while lunging to the left. 6. Repeat an equal number of reps on each side (6–8).

Odd position strength

For the strength training phase of a fascia-aware periodization program, our focus is on working with submaximal loads in odd positions at

length. The idea is to load odd positions at length so we can engage tissue paths across the entire body on all three planes. To be clear, we're not saying that shortening muscles under load is

Periodized fascia training for speed, power, and injury resilience



Figure 48.2 A-C Alternate Interior Reach

The alternate interior reach drill is another example of vector variability. It activates the core and engages fascia connections in a chain reaction along the transverse plane. **1.** Start by holding the load by the outside edges in an athletic stance. **2.** Rotate torso while reaching across the frontal plane as far as possible with each hand. **4.** Repeat an equal number of reps on each side (**6–8**).

bad and lengthening under load is good. They are both good, but they are different inputs that produce different results. From a fascia training perspective, the body needs to be subjected to a variety of different load paths on all three planes involving multiple joints, tissues, and structures. By lengthening muscles under load, shape stability and strength can be developed across the entire fabric of the body, not just over a local joint or segment.

Power and speed

In the world of athletic training, power is considered an expression of strength. If strength is being expressed at speed, it is often described in the industry as either speed-strength or strength-speed. While these terms are sometimes used interchangeably, they involve different training approaches and outcomes. Speed-strength refers to moving a load at a high rate of speed (e.g. a lineman in football pushing his opponent), while strength-speed is moving a relatively heavy load with the intention of

moving it as fast as possible (e.g. Olympic lifting). In the power development phase of periodized training, the focus is on movements that involve rapid oscillating speeds and rebound motions where the contact times are under 1.2 seconds (Kawakami et al., 2002). This means doing things like bouncing a medicine ball against a wall, jumping a rope, running on the balls of the feet, or quickly moving a submaximal load overhead from one shoulder to the other. When these inputs are done at a faster athletic pace, they will stimulate cell signaling to lay down new collagen in the fascia tissue. The elastic recoil power of collagen fiber is best exemplified by a kangaroo's astounding ability to jump 40-feet at a time in rapid succession and achieve speeds of up to 40-mph. Studies show that this impressive ability doesn't come from kangaroos having more fast twitch muscle fibers than other animals, as was originally thought (in fact, they have roughly the same amount of type II muscle fibers as koala bears). It comes from the stored potential energy in

Chapter 48

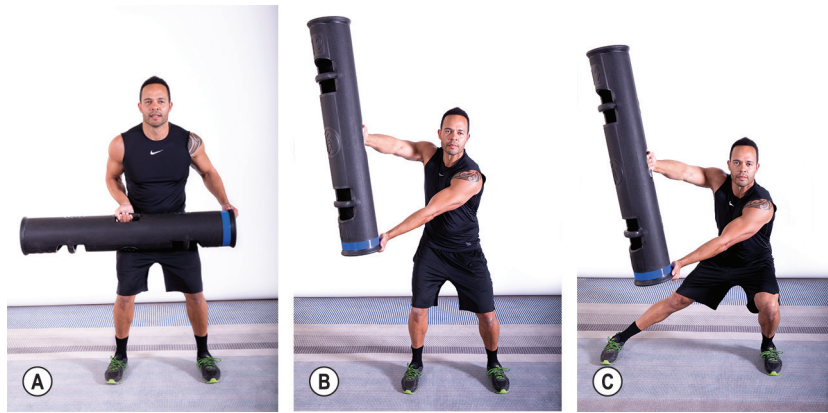


Figure 48.3 A–C Flag Lunge

The flag lunge helps develop odd-position strength by activating whole-body load paths and co-contractions in three dimensions on all three planes. Work with any loaded movement tool such as a ViPR PRO (pictured), medicine ball, or kettle bell to develop odd-position strength using the external load and gravity as drivers. **1.** Start by holding a load at midline with a shovel grip in an athletic stance. **2.** Raise the load laterally to one side, like a flag. **3.** Lunge across the frontal plane in the opposite direction of the load while exhaling forcefully. **4.** Return to the starting position. **5.** Repeat an equal number of repetitions on each side (**6–8**).

the collagen rich fascia tissue of their massive hindleg tendons and a pulsing elastic recoil dynamic called the “catapult effect” (Kram and Dawson, 1998). Humans are the only primates who have this same biological ability. The catapult effect is enabled by the surrounding muscles, which pre-contract isometrically to stretch the attached connective tissues—loading them like a stretched rubber band—and then relax to quickly release the stored elastic energy in an explosive pulse of force (Kawakami et al., 2002). The key to harnessing this powerful dynamic is the ability to create “super-stiffness” across the body’s integrated myofascial system. Studies conducted by Stuart McGill and Stephen Brown revealed that super-stiffness comes from multiple structures co-contracting simultaneously to create a “mechanical composite” of tissues that provide a pulse of rock-solid stability (Brown and McGill, 2009). These pulses of super-stiffness create the anchor points a kangaroo (or human) needs to rapidly store and release elastic energy relative to the density of

collagen fibers in their tendons. Therefore, a well-structured fascia-aware training program should start by developing a solid foundation of odd position strength with omnidirectional shape stability first. Power and speed can then be stacked on top.

When it comes to speed training, regardless of whether you’re training for linear speed or the rapid change of direction speeds experienced in field and court sports, it turns out that maximum velocity sprinting is unequaled as a training stimulus when prescribed in the right amounts. In fact, there is a growing body of evidence showing that maximum velocity sprinting not only improves overall speed and agility in short distances, it helps reduce soft tissue injuries in the groin and hamstring (Edouard et al., 2019).

Speed, agility, and quickness

As a running back in American football, the first line of defense is passed by cutting, darting, and

Periodized fascia training for speed, power, and injury resilience



Figure 48.4 A-B Power Bounds

Power bounds develop the ability to use the “catapult effect” of the tendons and fascia system. Powerful extension and flexion from the ankle, knee, hip, core and shoulder are needed to maximize force generation. This exercise forces the tendons and fascia system to engage in the high-intensity movement of loaded single-leg bounding. **Step 1:** Begin in a sprinter’s stance and start by exploding backward off one leg while driving the opposite knee forward and the opposite arm backward. **Step 2:** Let the ground come to you on each stride. Upon landing, immediately explode off the ground again into the next stride driving the opposite arm and leg backward. **Step 3:** Repeat the power bounds continuously while striving to cover more horizontal distance on each side. **Step 4:** Perform for 20–40 yards for 4–8 sets.

quickly changing directions. This rapid darting ability comes from the capacity to convert power into braking, accelerating, and rapidly reaccelerating in different directions. It requires the ability to instantly read visual and audio cues and quickly respond with appropriate counter-movements while controlling your momentum and body mass. It is a responsive motor skill that is reactive and unplanned. It therefore has a significant neurological component that is facilitated in part by the proprioceptive properties of the fascia system (Schleip, 2017). This means power-conversion drills that combine timing, balance, and posterior chain strength—like the speed-skater bound—are a crucial component in periodized speed training. Because they require refined movement literacy, neurological tuning,

and core strength. Rapid, whole-body compound movements prompt the body to utilize multiple fascial sling systems to absorb force, stabilize momentum, decelerate body mass, and produce explosive power using the catapult effect of the tendons working in conjunction with the myofascial system.

Rest and recovery

When it comes to rest and recovery, the traditional focus is on neural and metabolic recovery periods between workout sessions or circuits. However, there is a strong reason to monitor work-to-rest ratios over the course of each workout: this concerns the extra cellular matrix. For example, in a metabolically-intense CrossFit

Chapter 48



A

B

Figure 48.5 A-B Straight-Leg Bounding

Straight leg bounding is designed to activate the hamstrings and generate horizontal and vertical driving force off the ground. By keeping the leg straight and knees locked it forces the hamstrings to do more of the work as opposed to the quad and glut muscles. Since the hamstrings are a complex, two-jointed muscle responsible for hip extension and knee flexion it is commonly injured. Step 1: Start by standing tall with both legs firmly locked at the knee joint. Step 2: Flex and extend the hip in a shuffle like action while moving forward and slowly increase the intensity of the force production off the ground on each stride. Step 3: Stay leaning slightly forward with a strong active anterior core while syncing the arms with the legs to maximize force production into the ground. Step 4: Do two sets of 20–30 yards.

style workouts, injury rates tend to increase after about 30 minutes of exercise (Weisenthal et al., 2014). This is due to repetitive, aggressive muscle contractions that push blood and water away from localized areas of the body via muscle pumping and osmotic fluid pressure.

When water is pushed out of localized areas, the composition of the ECM is changed, and the fascia tissue becomes less dynamic and stiffness is reduced (Schleip and Müller, 2012). However, when water binds to a sugar receptor within fascia (i.e. it becomes bound water), the tissue

Periodized fascia training for speed, power, and injury resilience



Figure 48.6 A–C Speed-Skater Bounds

The speed skater bound is an example of speed-strength conversion. It is a total body movement with a focus on the gluteus maximus and posterior fascia chain. Due to the extreme hip flexion and stretch put on the glute complex while in this bent over position, the initial dynamic contraction challenges the body's fascia slings more than traditional exercises. It is an expression of power and control that relies on neural activity, proprioception, and timing. **Step 1:** Start standing on one leg flexed at the hip and knee with the opposite hand down in front of the grounded leg. **Step 2:** Dynamically and explosively jump laterally as fast as possible landing and balancing on the opposite leg. Focus on full extension of the hip and knee of the jumping leg while engaging in the lateral movement. **Step 3:** Immediately upon landing, jump back laterally as fast as possible. **Step 4:** Repeat continuous jumps for 12–16 total reps, 6–8 reps on each leg for 2–3 sets

becomes stiffer and more resistant to compression. Therefore, it is important to include strategic recovery sessions within workout routines, as opposed to purely between sessions. During an intense session, we recommend 10 minutes of rest for every 30 minutes of exhaustive exercise, using restorative pumping actions that return water and blood to the challenged tissues. This may enable water to bind to the fascia cells and sugar receptors, arguably creating a more robust, dynamic, and supportive fascial network throughout the workout.

In conclusion, the goal of periodized training is to engineer structural and functional adaptations in the body's tissues over time that improve athletic performance and resiliency. These adaptations are shown to be directly proportional to the mechanical stress inputs created by the

intensity (load), volume (quantity), tempo (rate), and frequency, of the training. With increasing challenge along omni-directional paths using varying loads at varying tempos—and enough time to recover and remodel—the tissues will adapt by becoming three-dimensionally stronger and more elastic along the lines of stress. Therefore, a fascia training periodization program should progress through a series of whole-body exercise routines that emphasize vector variability, then odd-position strength, then power, then speed, agility, and quickness—utilizing different loads, angles, and speeds throughout. By using this structure for a periodized program, the fascia system's natural biological behavior can potentially be utilized to make athletes stronger, faster, and bouncier, with less injury risk—which is the ultimate win.

Chapter 48



(A)



(B)

Figure 48.7 A-B Wide Outs

Wide-outs dynamically activate the gluteus medius/maximus and adductor group. By staying low and abducting then, immediately upon ground contact, adducting the legs in a jumping fashion while maintaining balance and a low athletic position, this exercise is a great lateral quickness drill. **Step 1:** Start in a wide stance squat position with knees in-line with your toes and hands behind your back.

Step 2: Without increasing your jump height, bring your feet close together by adducting the legs in a synchronized jump; then land softly maintaining a strong core without increasing the height of your head or hips. **Step 3:** Immediately upon landing, jump again abducting both legs and separating them back to the starting position. **Step 4:** Repeat the jumps continuously for two sets of 8–12 reps.

References

- Bohm, S., Mersmann, F., Tettke, M., Kraft, M. & Arampatzis, A. (2014) Human achilles tendon plasticity in response to cyclic strain: Effect of rate and duration. *Journal of Experimental Biology*. 217: 4010–4017
- Brown, S. & McGill, S. (2009) Transmission of muscularly generated force and stiffness between layers of the rat abdominal wall. *SPINE*. 34 (2).
- Clark, K.P. & Weyand, P.G. (2014) Are running speeds maximized with simple-spring stance mechanics? *Journal of Applied Physiology*. 117 (6).
- Edouard P., Mendiguchia, J., Guex, K., et al. (2019) Sprinting: a potential vaccine for hamstring injury? *Science Performance and Science Reports*.
- Ingber, D. (1998) The Architecture of Life. *Scientific American*.
- Järvinen, M.J. & Lehto, M.U.K. (1993) The effects of early mobilisation and immobilisation on the healing process following muscle injuries. *Sports Med*. 15(2): 78–89.
- Kawakami, Y., Muraoka, T., Ito, S., Kanehisa, H. & Fukunaga, T. (2002) In vivo muscle fiber behavior during

Periodized fascia training for speed, power, and injury resilience

- counter-movement exercise in humans reveals a significant role for tendon elasticity. *Journal of Physiology*. 540 (pt2): 635–646.
- Maas, M. & Sandercock, T.G. (2010) Force transmission between synergistic skeletal muscles through connective tissue. *Journal of Biomedicine and Biotechnology*. April 12.
- Kram, R. & Dawson, T. (1998) Energetics and biomechanics of locomotion by red kangaroos (*Macropus rufus*). *Comp Biochem Physiol*. 120 (1).
- Naito, K., Takagi, H. & Maruyama, T. (2011) Mechanical work, efficiency and energy redistribution mechanisms in baseball pitching. *Journal of Sports Technology*. 4 (1–2).
- Pollack, G.H. (2001) *Cells, Gels, and the Engines of Life. A New, Unifying Approach to Cell Function*. Seattle, Washington: Ebner and Sons Publishers.
- Sawicki, G. (2009) *It Pays to Have a Spring in Your Step*. Department of Ecology and Evolutionary Biology, Brown University.
- Schleip, R. & Müller, D.G. (2012) Training principles for fascial connective tissues: Scientific foundation and suggested practical applications. *Journal of Bodywork and Movement Therapies* 17 (1).
- Schleip, R. (2017) Fascia as a sensory organ: clinical applications. *Terra Rosa*. 20.
- Weisenthal, B.M., Beck, C.A., Maloney, M.D. et al. (2014) Injury Rate and Patterns Among CrossFit Athletes. *Orthop J Sports Med*. 2(4).
- Wiesinger, H.P., Rieder, F. et. al. (2017) Sport-specific capacity to use elastic energy in the patellar and achilles tendons of elite athletes. *Frontiers in Physiology*. 8 (132).
- Zügel, M. et al. (2018) Fascial tissue research in sports medicine: From molecules to tissue adaptation, injury and diagnostics. *British Journal of Sports Medicine*. 52 (23).